

MERJEM MEMIC

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Education

University of Michigan

Ann Arbor, MI

B.S. in Data Science and Computer Science

Expected Dec 2025

- **GPA:** 3.85/4.00 | **Awards:** Dean's List, Academic Scholarship
- **Courses:** Data Structures & Algorithms, Practical Data Science, Web Systems, Machine Learning, Linear Regression, Statistics and Probability, Natural Language Processing (NLP), Computer Vision
- **Activities:** Muslim Engineering Society (Professional Development Chair), Michigan Data Science Team (Member)

Technical Skills

Languages: Python, C++, Golang, Javascript, SQL, R, HTML, CSS

Technologies: React, Flask, AWS, PyTorch, Pandas, Plotly, MongoDB, Git, Tableau, Excel, Bash, Rest APIs, Firebase

Interests: Hiking through the mountains, enjoying the open air, and crocheting small animals

Experience

Ford Motor Company

Dearborn, Michigan

Software Engineer Intern

May – Jul 2025

- Modernize applications by supporting cloud migration initiatives through assisting in transitioning applications from on-premises infrastructure to Google Cloud Platform (GCP)
- Designed and developed multiple prototypes of an AI-powered RAG chatbot to assist with wireless issue troubleshooting before presenting it to leadership.
- Collaborate with team members to outline, develop, test, and implement automation workflows for account creation and deletion processes.
- Contribute to the development and enhancement of wireless security management infrastructure, focusing on application functionality and performance improvements
- Gained hands-on experience with CI/CD pipeline architecture and deployment practices, enhancing understanding of automated software delivery workflows.

University of Michigan

Ann Arbor, Michigan

Blue Lab Research Assistant

Jan 2025 – Present

- Perform sentiment analysis using the Bluesky Data API and NLP techniques to quantify a shift in attitudes of users.
- Handle large amounts of data on user activities through batch and stream processing to analyze repeated phrases

Computer Science Engineering Mentor

Sept 2024 – Present

- Facilitate weekly mentoring sessions for a group of 4+ mentees per semester, addressing technical and professional topics, contributing to a program that supports 30+ students each semester in their journeys through CS

EcoData

Ann Arbor Michigan

Frontend Developer

Jan – Apr 2024

- Designed an interactive page for a website to help individuals plan meals without excessive spending and food waste

Projects

Scalable Search Engine | Python, Javascript, Hadoop, MapReduce, Web Distributed Systems, Shell Scripting

- Built a scalable search engine similar to Google or Bing that returns ranked information based on typed input
- Engineered a multi-stage pipeline to generate a segmented inverted index, significantly improving query response time and search performance based on scraped Wikipedia information.
- Optimized system performance by profiling indexing stages and enabling dynamic scaling under high load conditions.

Classification of Wearable Sensors for Human Activity Recognition | Github Link | Python, Pytorch, Deep Learning

- Implemented a Support Vector Machine, Convolution Neural Network, and Recurrent Neural Network to classify a person's activities based on sensor data collected from a smartphone's embedded accelerometer and gyroscope.
- Optimized an SVM, achieving a 96% accuracy on test data by leveraging kernelization and hyperparameter tuning.

Insta485 - Client Side Dynamic Pages | Javascript, SQL, Flask, React, AWS, HTML, CSS, Web Development

- Created responsive interfaces using React to implement authentication, post interactions, infinite scroll, personalized user accounts, and real-time UI updates
- Emphasized modular code organization, component reuse, and responsive design for an intuitive user experience.
- Applied Flask Python frameworks to develop REST APIs and route requests, and processed them by generating dynamic data from SQLite Database

Euchre Interactive Card Game Interface | C++, Object-Oriented Programming, Algorithms

- Designed a C++ interface to allow users to play a game of euchre with simulated players executing moves via a programmed strategy through object-oriented programming
- Ensured quality assurance by iteratively testing throughout coding process to prevent avoidable compounded errors